

Tiffany

Michigan Department of Environmental Quality – Water Resources Division
WASTEWATER DISCHARGE PERMIT APPLICATION
 SECTION I – General Information

Section I shall be completed by all permit applicants. Instructions for completing Section I, Pages 1 and 2, are on Page 2 of the Appendix. To submit additional information, see Page ii, Item 3.

Water Resources Division Use Only Receipt #: <u>12-17955</u> Permit ID #: <u>95475.1</u>	Cashier Use Only: 37000-40535-9412-481000-00
--	---

PLEASE TYPE OR PRINT

1	NPDES PERMIT NUMBER MI0020257																
2. APPLICANT	Applicant Name City of Cadillac Address 200 N. Lake St. Address 2 or P.O. Box City Cadillac State MI ZIP Code 49601 Telephone (with area code) 231-775-2841 FAX (with area code) 231-775-2876 Applicant Web Site Address http://www.cadillac-mi.net																
3. FACILITY	Facility Name 1 Cadillac WWTP Facility Name 2 Facility Name 3 Street Address (Do not use a P.O. Box Number) 1121 Plett Rd. City Cadillac State MI ZIP Code 49601 Telephone (with area code) 231-775-2841 FAX (with area code) 231-775-2876 Facility Web Site Address http://www.cadillac-mi.net NP2 572802-10-1 03/22 60000 42203 9312 48100000 AY 2012 \$750																
4. CONTACTS	<table style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 25%; vertical-align: top;"> ☒ Application Contact ☒ Facility Contact ☒ Discharge Monitoring Reports ☒ Storm Water Billing ☒ Biosolids Billing ☒ NPDES Annual Billing </td> <td style="width: 30%; vertical-align: top;"> First Name Hamid Title Water Resources Supervisor Address 1 200 N. Lake St. City Cadillac State MI ZIP Code 49601 Telephone (with area code) 231-775-2841 Fax Number 231-775-2876 </td> <td style="width: 25%; vertical-align: top;"> Last Name Rahbarnoohi Business Address 2 City Cadillac State MI ZIP Code 49601 e-mail address hrahbarnoohi@cadillac-mi.net </td> <td style="width: 20%;"></td> </tr> <tr> <td style="vertical-align: top;"> ☐ Application Contact ☐ Facility Contact ☐ Discharge Monitoring Reports ☐ Storm Water Billing ☐ Biosolids Billing ☐ NPDES Annual Billing </td> <td style="vertical-align: top;"> First Name Title Address 1 City State ZIP Code Telephone (with area code) Fax Number </td> <td style="vertical-align: top;"> Last Name Business Address 2 City State ZIP Code e-mail address </td> <td></td> </tr> <tr> <td style="vertical-align: top;"> ☐ Application Contact ☐ Facility Contact ☐ Discharge Monitoring Reports ☐ Storm Water Billing ☐ Biosolids Billing ☐ NPDES Annual Billing </td> <td style="vertical-align: top;"> First Name Title Address 1 City State ZIP Code Telephone (with area code) Fax Number </td> <td style="vertical-align: top;"> Last Name Business Address 2 City State ZIP Code e-mail address </td> <td></td> </tr> </table>					☒ Application Contact ☒ Facility Contact ☒ Discharge Monitoring Reports ☒ Storm Water Billing ☒ Biosolids Billing ☒ NPDES Annual Billing	First Name Hamid Title Water Resources Supervisor Address 1 200 N. Lake St. City Cadillac State MI ZIP Code 49601 Telephone (with area code) 231-775-2841 Fax Number 231-775-2876	Last Name Rahbarnoohi Business Address 2 City Cadillac State MI ZIP Code 49601 e-mail address hrahbarnoohi@cadillac-mi.net		☐ Application Contact ☐ Facility Contact ☐ Discharge Monitoring Reports ☐ Storm Water Billing ☐ Biosolids Billing ☐ NPDES Annual Billing	First Name Title Address 1 City State ZIP Code Telephone (with area code) Fax Number	Last Name Business Address 2 City State ZIP Code e-mail address		☐ Application Contact ☐ Facility Contact ☐ Discharge Monitoring Reports ☐ Storm Water Billing ☐ Biosolids Billing ☐ NPDES Annual Billing	First Name Title Address 1 City State ZIP Code Telephone (with area code) Fax Number	Last Name Business Address 2 City State ZIP Code e-mail address	
☒ Application Contact ☒ Facility Contact ☒ Discharge Monitoring Reports ☒ Storm Water Billing ☒ Biosolids Billing ☒ NPDES Annual Billing	First Name Hamid Title Water Resources Supervisor Address 1 200 N. Lake St. City Cadillac State MI ZIP Code 49601 Telephone (with area code) 231-775-2841 Fax Number 231-775-2876	Last Name Rahbarnoohi Business Address 2 City Cadillac State MI ZIP Code 49601 e-mail address hrahbarnoohi@cadillac-mi.net															
☐ Application Contact ☐ Facility Contact ☐ Discharge Monitoring Reports ☐ Storm Water Billing ☐ Biosolids Billing ☐ NPDES Annual Billing	First Name Title Address 1 City State ZIP Code Telephone (with area code) Fax Number	Last Name Business Address 2 City State ZIP Code e-mail address															
☐ Application Contact ☐ Facility Contact ☐ Discharge Monitoring Reports ☐ Storm Water Billing ☐ Biosolids Billing ☐ NPDES Annual Billing	First Name Title Address 1 City State ZIP Code Telephone (with area code) Fax Number	Last Name Business Address 2 City State ZIP Code e-mail address															

Michigan Department of Environmental Quality – Water Resources Division
WASTEWATER DISCHARGE PERMIT APPLICATION
SECTION I – General Information

PLEASE TYPE OR PRINT

FACILITY NAME Cadillac WWTP	NPDES PERMIT NUMBER MI0020257
--------------------------------	----------------------------------

5. PERMIT ACTION REQUESTED (Check one box only). Instructions for this item are on Page 2 of the Appendix.

- ☐ **NEW USE.** A proposed discharge.
☐ **EXISTING DISCHARGE** that is currently unpermitted.
☒ **REISSUANCE** of current permit.
☐ **MODIFICATION** of current permit. Attach a description of the proposed modification.

Note: Applications for **New Use** discharges, **Existing Discharges** that are currently unpermitted, and for either **Reissuance** or **Modification** that include an increased loading of pollutants to the receiving water are required to submit a Rule 98 Demonstration with the Application. See Item 6.

6. RULE 98 – ANTIDEGRADATION REQUIREMENTS. Instructions for this item are on Page 2 of the Appendix.

In accordance with Rule 323.1098 of the Michigan Water Quality Standards, the applicant is required to submit an Antidegradation Demonstration for any new or increased loading of pollutants to the surface waters of the state. An Antidegradation Demonstration must contain the information specified in Rule 1098, outlined on Pages 8-9 of the Appendix. For assistance in completing this item, contact the Permits Section.

Will this discharge be an increased loading of pollutants to the surface waters of the state? ☐ Yes, continue below. ☒ No.

- ☐ Antidegradation Demonstration provided. ☐ Increased loading of pollutants is exempt from Antidegradation Demonstration as indicated below:
- ☐ A short-term (weeks to months) or temporary lowering of water quality
 - ☐ Bypasses that are not prohibited by regulations set forth in 40 CFR 122.41(m)
 - ☐ Response actions undertaken to alleviate a release of pollutants into the environment that may pose an imminent and substantial danger to the public health or welfare
 - ☐ Discharges of pollutant quantities from the intake water at a facility if the intake and discharge are to the same body of water
 - ☐ Increases in flow at a POTW if the increase is within the design flow of the facility, there is no increased loading of BCCs that are not specifically limited in the current permit, and there is no significant change expected in the characteristics of the wastewater collected
 - ☐ Intermittent increased loading related to wet-weather conditions
 - ☐ New or increased loading due to DEQ-approved controls related to wet-weather conditions
 - ☐ Discharges authorized by Certificates of Coverage (COC) and Notices of Coverage
 - ☐ Increased loadings within the authorized levels of a limit in an existing control document, except those loadings that result from actions by the permittee that would otherwise require submittal of an increased use request
 - ☐ Increased loadings of a pollutant which do not involve Bioaccumulative Chemicals of Concern and which use less than 10 percent of the unused loading capacity that exists at the time of the request

7. ADDITIONAL FACILITY LOCATION INFORMATION. Instructions for this item are on Page 2 of the Appendix.

A	Local Unit of Government (LUG) City of Cadillac	LUG e-mail address utilities@cadillac-mi.net												
B	County Wexford	Township Haining												
C	<table style="width: 100%; border: none;"> <tr> <td style="width: 15%;">Town</td> <td style="width: 15%;">Range</td> <td style="width: 15%;">Section</td> <td style="width: 15%;">¼</td> <td style="width: 15%;">¼, ¼</td> <td style="width: 30%;">Private (French) Land Claim</td> </tr> <tr> <td>22N</td> <td>9W</td> <td>33</td> <td>NE</td> <td>NE</td> <td></td> </tr> </table>	Town	Range	Section	¼	¼, ¼	Private (French) Land Claim	22N	9W	33	NE	NE		
Town	Range	Section	¼	¼, ¼	Private (French) Land Claim									
22N	9W	33	NE	NE										
D	Latitude N 44 15.118	Longitude W 085 23.856												

8. CERTIFIED OPERATOR

Does the facility have a DEQ-certified operator? ☒ Yes ☐ No Instructions for this item are on Page 2 of the Appendix.

First Name Hamid	Last Name Rahbarnoohi		
Certification Number 11717	Certification Classification(s) B, C, and D		
Address 1 1121 Plett Rd.	Address 2		
City Cadillac	<table style="width: 100%; border: none;"> <tr> <td style="width: 40%;">State MI</td> <td style="width: 60%;">Zip Code 49601</td> </tr> </table>	State MI	Zip Code 49601
State MI	Zip Code 49601		
Telephone Number 231-775-2841	<table style="width: 100%; border: none;"> <tr> <td style="width: 40%;">Fax Number 231-775-2876</td> <td style="width: 60%;">e-mail address hrahbarnoohi@cadillac-mi.net</td> </tr> </table>	Fax Number 231-775-2876	e-mail address hrahbarnoohi@cadillac-mi.net
Fax Number 231-775-2876	e-mail address hrahbarnoohi@cadillac-mi.net		

Michigan Department of Environmental Quality – Water Resources Division
WASTEWATER DISCHARGE PERMIT APPLICATION
SECTION I – General Information

PLEASE TYPE OR PRINT

FACILITY NAME Cadillac WWTP	NPDES PERMIT NUMBER MI0020257
--------------------------------	----------------------------------

9. OTHER ENVIRONMENTAL PERMITS

Provide the information requested below for any other federal, state, or local environmental permits in effect or applied for at the time of submittal of this Application, including, but not limited to, permits issued under any of the following programs: Air Pollution Control, Hazardous Waste Management, Wetlands Protection, Soil Erosion and Sedimentation Control, and other NPDES permits. To submit additional information, see Page ii, Item 3.

Issuing Agency	Permit or COC Number	Permit Type

10. WATER FLOW DIAGRAM AND NARRATIVE DESCRIPTION

Provide a flow diagram (using 8½" x 11" paper if possible) and a narrative description that explains the diagram. The diagram should show the wastewater flow through the facility (from intake through discharge), including all processes, treatment units, including any lagoons or ponds used for wastewater treatment or storage (identify treatment units that operate intermittently), and bypass piping. Show all operations contributing wastewater and the locations of flow meters, chemical feeds, and monitoring and discharge points. The water balance shall show the daily average flow rates at the intake and discharge points, and approximate daily flow rates between treatment units, including influent and treatment rates. Use actual measurements whenever available, otherwise use the best estimate. Show all significant losses of water to products, atmosphere, and discharge. In addition, provide a flow diagram for any storm water discharges from secondary structures that are required by state or federal law and for storm water runoff from any Site of Environmental Contamination, pursuant to Part 201 of the Michigan Act. **Do not send blueprints. Provide black-and-white reproducible diagrams.**

Municipal Facilities – Include a narrative that briefly describes the history of the wastewater treatment facility and collection system, including the initial construction, facility improvements, future plans for upgrade, location of all constructed emergency overflows, and other pertinent information.

Industrial and Commercial Facilities – The diagram shall include all operations contributing wastewater, including process and production areas, sanitary flows, cooling water, and storm water runoff. Include a narrative that provides a brief description of the nature of the business and the manufacturing processes.

ATTACH THIS INFORMATION TO THIS APPLICATION. PLEASE DO NOT BIND THIS INFORMATION. Comments:

11. MAP OF FACILITY AND DISCHARGE LOCATION

Provide a detailed black-and-white reproducible map on 8½" x 11" paper showing the location of the existing or proposed facility, wastewater and biosolids treatment system(s), water intakes, wastewater monitoring, and wastewater discharge points into receiving waters (including bypasses). Include the exact location of the water intakes, wastewater monitoring and discharge point(s) and, if applicable, all areas through which the discharge flows (e.g., wetlands, open drains, storm sewers) between the discharge point and the receiving water. If the discharge is to a storm sewer, label the storm sewer and show its flow path to the receiving water. Also include the location of any water supply intakes or wells and groundwater monitoring wells. This map shall be a United States Geological Survey quadrangle (7.5 minute series) or other map of comparable detail, scale, and quality (which shows surface water bodies, roads, bathing beaches, and other pertinent landmarks). **It is preferred that the minimum area this map shall encompass be approximately one (1) mile beyond the property boundaries.**

ATTACH THIS INFORMATION TO THIS APPLICATION. Comments:

Michigan Department of Environmental Quality – Water Resources Division
WASTEWATER DISCHARGE PERMIT APPLICATION
SECTION I – General Information

PLEASE TYPE OR PRINT

FACILITY NAME Cadillac WWTP				NPDES PERMIT NUMBER MI0020257	
12. CONTRACT LABORATORIES THAT PROVIDE ANALYTICAL SUPPORT Provide the name and address of each contract laboratory or consulting firm that performed any analyses submitted as part of this Application. To submit additional information, see Page ii, Item 3.					
Laboratory Name KAR laboratories, Inc.			Laboratory Name		
Street Address 4425 manchester Rd.			Street Address		
City Kalamazoo	State MI	ZIP Code 49001	City	State	ZIP Code
Telephone (with area code) 269-381-9666		Fax (with area code) 269-381-9698		Telephone (with area code)	
Analysis Performed Mercury, Metals, Lindane, Organics, and Biosolids		Analysis Performed			
Laboratory Name Paragon Laboratories			Laboratory Name		
Street Address 12649 Richfield Ct.			Street Address		
City Livonia	State MI	City 48150	State	City	State
Telephone (with area code) 734-462-3900		Fax (with area code) 734-462-3911		Telephone (with area code)	
Analysis Performed Wet Test		Analysis Performed			
13. LIST ADJACENT PROPERTY OWNERS List the names and mailing addresses of all property owners for all properties adjacent to the facility, treatment systems, and discharge locations. For vacant lots or empty buildings, supply the owner's mailing address – NOT the lot or building property address. To submit additional information, see Page ii, Item 3.					
Name	Address	City	State	ZIP Code	
David Pierson	1109 Plett Rd.	Cadillac	MI	49601	
Kelly Cherette	1118 Sunberg Rd.	Cadillac	M	49601	
Dennis Philips & Sheila Gullekson	1117 Sunberg Rd.	Cadillac	MI	49601	
Mark Darrow	1106 Ernst St.	Cadillac	MI	49601	
William and Helen Heath	1110 Plett Rd.	Cadillac	MI	49601	
Gordon Maxwell	1107, 1109, 1115 Wheeler St.	Cadillac	MI	49601	
Joan Padgett	1208 Riverside St.	Cadillac	MI	49601	
Andrew Maler & Rita Smith	1204 Riverside St.	Cadillac	MI	49601	
Gordon Maxwell	1200 and 1202 Riverside St.	Cadillac	MI	49601	
Jami and Brandy Loveless	1138 Riverside St.	Cadillac	MI	49601	
Ben Bigelow	1134 Riverside St.	Cadillac	MI	49601	
Albin and Barbara Johnson	1132 Riverside St.	Cadillac	MI	49601	

PLEASE TYPE OR PRINT

List the names and mailing addresses of all property owners for all properties adjacent to the facility, treatment systems, and discharge locations. For vacant lots or empty buildings supply the owners mailing address not the lot or building property address. To submit additional information, see Page ii, Item 3.

EQP 4659-A (Rev. 1/2006)

Michigan Department of Environmental Quality – Water Resources Division
WASTEWATER DISCHARGE PERMIT APPLICATION
SECTION I – General Information

PLEASE TYPE OR PRINT

FACILITY NAME

Cadillac WWTP

NPDES PERMIT NUMBER

MI0020257

14. APPLICATION CERTIFICATION

Rule 323.2114(1-4), promulgated under the Michigan Act, requires that this Application must be signed as follows:

- A. For an organization, company, corporation, or authority, by a principal executive office, vice president, or higher
- B. For a partnership, by a general partner
- C. For a sole proprietor, by the proprietor
- D. For a municipal, state, or other public facility, by a principal executive officer or ranking elected official (e.g., mayor, village president, city or village manager, or clerk)

Note: If the signatory is not listed above, but is authorized to sign the Application, please provide documentation of that authorization.

"I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system, or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for having knowledge of violations."

The last Application for this facility was submitted on: 03/26/2007

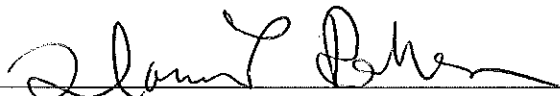
I understand that my signature constitutes a legal agreement to comply with the requirements of the NPDES Permit. I certify under penalty of law that I possess full authority on behalf of the legal owner/permittee to sign and submit this Application.

Print Name HAMID RAHBARNOOHI

Title

WATER RESOURCES SUPERVISOR

Signature



Date

03/05/2012

This completes Section I. Publicly-Owned Treatment Works discharging sanitary and industrial wastewater to the surface waters, and privately-owned treatment works discharging sanitary wastewater to the surface waters should complete Section II. Privately-owned treatment works include, but are not limited to, Mobile Home Parks, Campgrounds, Condominiums, Hotels and Motels, and Nursing Homes. All other applicants should complete Section III. If assistance is needed to complete this Application, contact the Permits Section.

Permit Application Submittal Checklist

Please confirm the following before submitting the Application:

- ☒ 1. Section I has been completed, including all diagrams, maps, and the treatment process narrative.
- ☒ 2. The Application has been signed as required above in Section I.14.A.-D. or a copy of the letter authorizing the signatory to sign the letter has been included, as appropriate.
- ☒ 3. Section II or Section III has been completed, including any additional information or submissions.
- ☒ 4. Section IV has been completed by any facility that discharges storm water.
- ☐ 5. Section V has been completed by any facility that is a Concentrated Animal Feeding Operation.
- ☐ 6. Section VI has been completed by any facility that has Cooling Water Intake Structures.
- ☒ 7. A check or money order for the appropriate application fee has been made out to the "State of Michigan" and has been included with the Application submittal.
- ☒ 8. E-mail addresses have been provided.

Michigan Department of Environmental Quality – Water Resources Division
WASTEWATER DISCHARGE PERMIT APPLICATION
SECTION II – Sanitary Wastewater

Section II is to be completed by Publicly-Owned Treatment Works (POTWs) discharging treated or untreated sanitary and industrial wastewater to the surface waters. Section II is also to be completed by all privately-owned treatment works discharging treated sanitary wastewater to the surface waters. The privately-owned treatment works include, but are not limited to, Mobile Home Parks, Campgrounds, Condominiums, Hotels and Motels, and Nursing Homes.

A. Facility Information

PLEASE TYPE OR PRINT

FACILITY NAME Cadillac WWTP	NPDES PERMIT NUMBER MI0020257
--------------------------------	----------------------------------

1. WATER SUPPLY INFORMATION

List the source(s) of the water supply in the area served by sewers. Identify groundwater wells and surface water intakes, as well as the name(s) of any surface water(s) from which intake water is drawn.

2. SERVICE AREA INFORMATION

POTWs are required to provide the following information: List the governmental jurisdictions (e.g., cities, townships, villages) that this facility serves (applicants should include themselves). What is the population in each jurisdiction? Is the jurisdiction's collection system separate, combined, or both? If the collection system is both separate and combined, what percentage is combined? To submit additional information, see Page ii, Item 3.

Municipality and E-Mail Address	Type of Collection System	Percent Combined	Population Served
City of Cadillac http://www.cadillac-mi.net	☒ Separate ☐ Combined	0	10500
Hairing Township http://haringtwp.org/	☒ Separate ☐ Combined	0	700
Cherry Grove Township http://www.cherrygrovetwp.org/	☒ Separate ☐ Combined	0	1000
Clam Lake Township http://www.clamlaketownship.org/	☒ Separate ☐ Combined	0	500
_____	☒ Separate ☐ Combined	_____	_____

Total population served by this facility: 12700

Privately-Owned Treatment Works are required to provide the following information:

Describe the area served by this facility (e.g., mobile home park, condominium, nursing home).

Provide the number of residential units served by this facility: _____

3. BIOMONITORING FOR ACUTE AND CHRONIC TOXICITY

POTWs meeting one or more of the following criteria are required to submit with this Application the results of four (4) Whole Effluent Toxicity (WET) tests for each of the facility's discharge points, excluding combined sewer overflows: 1) POTWs with a design flow rate greater than or equal to one (1) million gallons per day (MGD); 2) POTWs with an approved Federal Industrial Pretreatment Program (FIPP); and/or 3) POTWs required to develop a FIPP.

The results of the tests shall be reported using the Acute Toxicity Test Report, *Ceriodaphnia dubia* Chronic Toxicity Test Report, and the Fathead Minnow Chronic Toxicity Test Report available in the Appendix. Please do not submit additional forms or paperwork pertaining to WET tests with this Application.

At a minimum, the applicant shall submit the results of quarterly WET testing for a 12-month period prior to this Application or the results of annual WET tests conducted during the five years prior to this Application. In addition, the applicant shall submit the results of any other WET tests from the past five years. If a WET test in the past 4½ years revealed toxicity, provide all the information on the cause of toxicity or the results of all toxicity reduction evaluations, if any were conducted. The applicant does not need to submit results for previously-submitted WET tests. For assistance, see "Whole Effluent Toxicity Test Guidance and Requirements" on Page 17 of the Appendix. Comments:

Michigan Department of Environmental Quality – Water Resources Division
WASTEWATER DISCHARGE PERMIT APPLICATION
 SECTION II – Sanitary Wastewater

B. Outfall Information

Complete a separate Section II.B. Outfall Information (Pages 7 – 12) for each outfall at the facility. Make copies of Section II.B. for each additional outfall that discharges treated wastewater.

PLEASE TYPE OR PRINT

FACILITY NAME Cadillac WWTP	NPDES PERMIT NUMBER MI0020257	OUTFALL NUMBER 001
--------------------------------	----------------------------------	-----------------------

1. OUTFALL INFORMATION. Instructions for this item are on Page 3 of the Appendix.

A.	Receiving Water Clim River				Hydrologic Unit Code (HUC) 04060102	
B.	County Wexford				Township Haring	
C.	Town 22N	Range 09W	Section 33	1/4 NE	1/4, 1/4 NE	Private (French) Land Claim
D.	Latitude N44 15.949				Longitude W085 23.869	

E. Facility Annual Average Design Flow:

Seasonal Discharge: _____ MGY (Continue with Item F.) Continuous Discharge: 3.2 MGD (Continue with Item G.)

F. Seasonal Discharge:

List the discharge periods (by month) in the spaces provided below.

From	Through	Actual Discharge Volume (MGD)	Annual Total
From	Through	Actual Discharge Volume (MGD)	
From	Through	Actual Discharge Volume (MGD)	
From	Through	Actual Discharge Volume (MGD)	
From	Through	Actual Discharge Volume (MGD)	

G. Continuous Discharge:

How often is there a discharge from this outfall (on average)? 24 Hours/Day 365 Days/Year

Provide the actual facility flows for the past three years.

	Three Years Ago	Two Years Ago	Last Year
Annual Average Daily Flow (MGD)	2.025	2.108	2.140
Maximum Daily Flow in a Single Day (MGD)	2.689	2.743	3.259

Batch dischargers are required to provide the following additional information:

Is there effluent flow equalization? ☐ Yes ☐ No

Batch Peak Flow Rate: _____ Number of batches discharged per day: _____

	Minimum	Average	Maximum
Batch Volume (gallons)			
Batch Duration (minutes)			

H. Inflow and Infiltration:

What is the current average daily volume of inflow and infiltration at this outfall? _____ Gallons/Day

What corrective actions are being taken to minimize this inflow and infiltration?

Michigan Department of Environmental Quality – Water Resources Division
WASTEWATER DISCHARGE PERMIT APPLICATION
 SECTION II – Sanitary Wastewater

B. Outfall Information

PLEASE TYPE OR PRINT

FACILITY NAME Cadillac WWTP	NPDES PERMIT NUMBER MI0020257	OUTFALL NUMBER 001
--------------------------------	----------------------------------	-----------------------

2. EFFLUENT CHARACTERISTICS – CONVENTIONAL POLLUTANTS

Existing Treatment Works Treating Domestic Sewage (TWTDS) are required to report data from effluent sampled and analyzed by the permittee for the parameters listed below. (See the Definition Section on Page 8 in the Appendix for sampling definitions, including "maximum daily concentration" and "maximum monthly concentration.") Retention Treatment Basins are required to provide a summary of influent and effluent data for the last three years. For analytical test requirements, or if alternate test procedures for any parameter listed below have been approved, see Page ii, Item 5. If the data was previously submitted via DMRs, check the box and proceed to Item 3.

New TWTDS are required to provide estimated effluent concentrations for the parameters listed below.

☐ Check this box if additional information is included as an attachment. To submit additional information, see Page ii, Item 3.

Please Note: Rule 323.1062 allows the use of either *Escherichia coli* or Fecal Coliform Bacteria as the indicator that effluent has been disinfected. The DEQ will use the indicator selected below in the permit issued based on this Application.

☐ Use *Escherichia coli* as an indicator of disinfection.

☒ Use Fecal Coliform Bacteria as an indicator of disinfection.

Submitted via DMRs or e-DMRs	Parameter	Maximum Monthly Concentration	Maximum Daily Concentration	Units	Number of Analyses	Sample Type
☐	Biochemical Oxygen Demand – 5 day (BOD ₅)			mg/l		☐ Grab ☐ 24-Hr Comp
☐	BOD ₅ , Lowest % Removed		Do Not Use	%		☐ Grab ☐ 24-Hr Comp
☒	Carbonaceous BOD ₅ (CBOD ₅)			mg/l		☐ Grab ☒ 24-Hr Comp
☒	Carbonaceous BOD ₅ , Lowest % Removed		Do Not Use	%		☐ Grab ☒ 24-Hr Comp
☒	Ammonia Nitrogen (as N)			mg/l		☐ Grab ☒ 24-Hr Comp
☒	Total Suspended Solids			mg/l		☐ Grab ☒ 24-Hr Comp
☒	Total Suspended Solids, Lowest % Removed		Do Not Use	%		☐ Grab ☒ 24-Hr Comp
☐	Total Dissolved Solids			mg/l		☐ Grab ☐ 24-Hr Comp
☒	Total Phosphorus (as P)			mg/l		☐ Grab ☒ 24-Hr Comp
☒	Fecal Coliform Bacteria (report geometric means)		Maximum 7-day	counts/100 ml		Grab
☐	<i>Escherichia coli</i> (report geometric means)		Maximum 7-day	counts/100 ml		Grab
☐	Total Residual Chlorine			☐ µg/l ☐ mg/l		Grab
☒	Dissolved Oxygen	Do Not Use	Minimum Daily	mg/l		☒ Grab ☐ 24-Hr Comp
☒	pH	Minimum	Maximum	standard units		☒ Grab ☐ 24-Hr Comp
☒	Temperature			☒ °F ☐ °C		☒ Grab ☐ 24-Hr Comp
☐						☐ Grab ☐ 24-Hr Comp

Michigan Department of Environmental Quality – Water Resources Division
WASTEWATER DISCHARGE PERMIT APPLICATION
 SECTION II – Sanitary Wastewater

B. Outfall Information

PLEASE TYPE OR PRINT

FACILITY NAME Cadillac WWTP	NPDES PERMIT NUMBER MI0020257	OUTFALL NUMBER 001
--------------------------------	----------------------------------	-----------------------

3. EFFLUENT CHARACTERISTICS – TOXIC POLLUTANTS

Existing POTWs with (1) a design flow greater than or equal to 1.0 MGD; or (2) an approved Federal Industrial Pretreatment Program (FIPP); or (3) required to develop a FIPP or otherwise required by the permitting authority, shall provide the results of a minimum of three effluent analyses for each parameter listed below for each outfall through which effluent is discharged. Any effluent testing data for pollutants not specifically listed shall be submitted on separate pages. Do not include information on combined sewer overflows in this section.

All existing POTWs (unless already included above) are required to provide (1) the results of at least one effluent analysis (taken in the last three years) for any chemical that is known or believed to be present in facility effluent that is listed in Tables 2, 3, and 4 of the Appendix; (2) a measured or estimated effluent concentration for any chemical that is known or believed to be present that is listed in Table 5 of the Appendix; (3) a measured or estimated concentration for any toxic or otherwise injurious chemical known or believed to be present in facility effluent that is not previously identified in this Application; and (4) results of all other effluent analyses that have been performed within the past five years for any chemical listed in Tables 2, 3, 4, and 5 of the Appendix.

New POTWs are required to provide an estimated effluent concentration for any chemical expected to be present in facility effluent that is listed in Tables 2, 3, 4, and 5 of the Appendix, and an estimated effluent concentration for any toxic or otherwise injurious chemical known or believed to be present in facility effluent that has not been previously identified in this Application.

Note: If the effluent concentrations are estimated, place an E in the "Analytical Method" column. In accordance with Rule 323.1211(7), facilities whose supply water contains toxic pollutants that are withdrawn from and discharged to the same body of water may qualify for intake credits for those toxic pollutants. See Rule 1211(7) for qualification and demonstration requirements. Effluent data submitted in response to this part may be recorded on Pages 9 – 12, or by submission of sampling analytical reports. To submit additional information, see Page ii, Item 3. Report all sampling results in µg/l.

For analytical test requirements, or if alternate test procedures for any parameter listed below have been approved, see Page ii, Item 5 and Table 7 in the Appendix.

Submitted via DMRs or e-DMRs	SAMPLE DATE →		08/12/09	05/04/10	03/01/11	10/03/11		
	PARAMETER	CAS No.	Conc. (µg/l)	Conc. (µg/l)	Conc. (µg/l)	Conc. (µg/l)	Sample Type	Analytical Method
METALS (TOTAL RECOVERABLE), CYANIDE, PHENOLS, AND HARDNESS								
☐	Antimony	7440360	<1	<1	<1	<1	Comp	200.8
☐	Arsenic	7440382	2	<1	2	1	Comp	200.8
☐	Barium	7440393	16	18	15	20	Comp	200.8
☐	Beryllium	7440417	<1	<1	<1	<1	Comp	200.8
☐	Boron	7440428	355	220	182	216	Comp	200.7
☐	Cadmium	7440439	<0.2	<0.2	<0.2	<0.2	Comp	200.8
☐	Chromium	7440473	77.6	<5	<5	<5	Comp	200.8
☐	Copper	7440508	<5	8	15	8	Comp	200.8
☐	Lead	7439921	<1	<1	<1	<1	Comp	200.8
☒	Mercury (USEPA Method 1631)	7439976	0.0047	0.00075	0.00055	<0.0005	Comp	200.8
☐	Nickel	7440020	5	3	<5	5	Comp	200.8
☒	Selenium	7782492	<5	<2	<2	<2	Comp	200.8
☐	Silver	7440224	<0.5	<0.5	<0.5	<0.5	Comp	200.8
☐	Thallium	7440280	<2	<2	<1	<1	Comp	200.8
☒	Zinc	7440666	<57.8	<50	<87.5	<59.8	Comp	200.8
☒	Available Cyanide (Method OIA 1677)	57125	<2	<2	<2	<2	Grab	200.8
☐	Total Phenolic Compounds	None	<20	<20	<20	<20	Grab	200.8
☐	Hardness (as CaCO ₃)	None	270	224	223	217	Comp	130.2

Michigan Department of Environmental Quality – Water Resources Division
WASTEWATER DISCHARGE PERMIT APPLICATION
 SECTION II – Sanitary Wastewater

B. Outfall Information

PLEASE TYPE OR PRINT

FACILITY NAME Cadillac WWTP			NPDES PERMIT NUMBER MI0020257				OUTFALL NUMBER 001	
Submitted via DMRs or e-DMRs	SAMPLE DATE →		8/12/09	05/04/10	03/01/11	10/03/11		
	PARAMETER	CAS No.	Conc. (µg/l)	Conc. (µg/l)	Conc. (µg/l)	Conc. (µg/l)	Sample Type	Analytical Method
VOLATILE ORGANIC COMPOUNDS								
☐	Acrolein	107028	<20	<20	<20	<20	Grab	EPA 624
☐	Acrylonitrile	107131	<2	<2	<2	<2	Grab	EPA 624
☐	Benzene	71432	<1	<1	<1	<1	Grab	EPA 624
☐	Bromoform	75252	<1	<1	<1	<1	Grab	EPA 624
☐	Carbon Tetrachloride	56235	<1	<1	<1	<1	Grab	EPA 624
☐	Chlorobenzene	108907	<1	<1	<1	<1	Grab	EPA 624
☐	Chlorodibromomethane	124481	<1	<1	<1	<1	Grab	EPA 624
☐	Chloroethane	75003	<1	<1	<1	<1	Grab	EPA 624
☐	2-chloro-ethylvinyl ether	110758	<10	<10	<10	<10	Grab	EPA 624
☐	Chloroform	67663	<1	<1	<1	<1	Grab	EPA 624
☐	Dichlorobromomethane	75274	<1	<1	<1	<1	Grab	EPA 624
☐	1,1-dichloroethane	75343	<1	<1	<1	<1	Grab	EPA 624
☐	1,2-dichloroethane	107062	<1	<1	<1	<1	Grab	EPA 624
☐	Trans-1,2-dichloroethylene	156605	<1	<1	<1	<1	Grab	EPA 624
☐	1,1-dichloroethylene	75354	<1	<1	<1	<1	Grab	EPA 624
☐	1,2-dichloropropane	78875	<1	<1	<1	<1	Grab	EPA 624
☐	1,3-dichloropropylene	542756	<1	<1	<1	<1	Grab	EPA 624
☐	Ethylbenzene	100414	<1	<1	<1	<1	Grab	EPA 624
☐	Methyl Bromide	74839	<1	<1	<1	<1	Grab	EPA 624
☐	Methyl Chloride	74873	<1	<1	<1	<1	Grab	EPA 624
☐	Methylene Chloride	75092	<1	<1	<1	<1	Grab	EPA 624
☐	1,1,2,2-tetrachloroethane	79345	<1	<1	<1	<1	Grab	EPA 624
☐	Tetrachloroethylene	127184	<1	<1	<1	<1	Grab	EPA 624
☐	Toluene	108883	<1	<1	<1	<1	Grab	EPA 624
☐	1,1,1-trichloroethane	71556	<1	<1	<1	<1	Grab	EPA 624
☐	1,1,2-trichloroethane	79005	<1	<1	<1	<1	Grab	EPA 624
☐	Trichloroethylene	79016	<1	<1	<1	<1	Grab	EPA 624
☐	Vinyl Chloride	75014	<1	<1	<1	<1	Grab	EPA 624

Michigan Department of Environmental Quality – Water Resources Division
WASTEWATER DISCHARGE PERMIT APPLICATION
 SECTION II – Sanitary Wastewater

B. Outfall Information

PLEASE TYPE OR PRINT

FACILITY NAME Cadillac WWTP			NPDES PERMIT NUMBER MI0020257				OUTFALL NUMBER 001	
Submitted via DMRs or e-DMRs	SAMPLE DATE →		8/12/09	05/04/10	03/01/11	10/03/11		
	PARAMETER	CAS No.	Conc. (µg/l)	Conc. (µg/l)	Conc. (µg/l)	Conc. (µg/l)	Sample Type	Analytical Method
☐	P-chloro-m-cresol	None	<5	<5	<5	<5	Comp	EPA 625
☐	2-chlorophenol	95578	<5	<5	<5	<5	Comp	EPA 625
☐	2,4-dichlorophenol	120832	<5	<5	<5	<5	Comp	EPA 625
☐	2,4-dimethylphenol	105679	<5	<5	<5	<5	Comp	EPA 625
☐	4,6-dinitro-o-cresol	534521	<20	<20	<20	<20	Comp	EPA 625
☐	2,4-dinitrophenol	51285	<20	<20	<20	<20	Comp	EPA 625
☐	2-nitrophenol	88755	<5	<5	<5	<5	Comp	EPA 625
☐	4-nitrophenol	100027	<20	<20	<20	<20	Comp	EPA 625
☐	Pentachlorophenol	87865	<5	<5	<5	<5	Comp	EPA 625
☐	Phenol	108952	<5	<5	<5	<5	Comp	EPA 625
☐	2,4,6-trichlorophenol	88062	<5	<5	<5	<5	Comp	EPA 625
BASE-NEUTRAL COMPOUNDS								
☐	Acenaphthene	83329	<5	<5	<5	<5	Comp	EPA 625
☐	Acenaphthylene	208968	<5	<5	<5	<5	Comp	EPA 625
☐	Anthracene	120127	<5	<5	<5	<5	Comp	EPA 625
☐	Benzidine	92875	<50	<50	<50	<50	Comp	EPA 625
☐	Benzo(a)anthracene	56553	<5	<5	<5	<5	Comp	EPA 625
☐	Benzo(a)pyrene	50328	<5	<5	<5	<5	Comp	EPA 625
☐	3,4 benzofluoranthene	205992	<5	<5	<5	<5	Comp	EPA 625
☐	Benzo(ghi) perylene	191242	<5	<5	<5	<5	Comp	EPA 625
☐	Benzo(k)fluoranthene	207089	<5	<5	<5	<5	Comp	EPA 625
☐	Bis (2-chloroethoxy) methane	111911	<5	<5	<5	<5	Comp	EPA 625
☐	Bis (2-chloroethyl) ether	111444	<5	<5	<5	<5	Comp	EPA 625
☐	Bis (2-chloroiso-propyl) ether	108601	<5	<5	<5	<5	Comp	EPA 625
☐	Bis (2-ethylhexyl) phthalate	117817	<5	<5	<5	<5	Comp	EPA 625
☐	4-bromophenyl phenyl ether	101553	<5	<5	<5	<5	Comp	EPA 625
☐	Butyl benzyl phthalate	85687	<5	<5	<5	<5	Comp	EPA 625
☐	2-chloronaphthalene	91587	<5	<5	<5	<5	Comp	EPA 625
☐	4-chlorophenylphenyl ether	7005723	<5	<5	<5	<5	Comp	EPA 625

Michigan Department of Environmental Quality – Water Resources Division
WASTEWATER DISCHARGE PERMIT APPLICATION
 SECTION II – Sanitary Wastewater

B. Outfall Information

PLEASE TYPE OR PRINT

FACILITY NAME Cadillac WWTP			NPDES PERMIT NUMBER MI0020257				OUTFALL NUMBER 001	
Submitted via DMRs or e-DMRs	SAMPLE DATE →		8/12/09	05/04/10	03/01/11	10/03/11	Sample Type	Analytical Method
	PARAMETER	CAS No.	Conc. (µg/l)	Conc. (µg/l)	Conc. (µg/l)	Conc. (µg/l)		
☐	Chrysene	218019	<5	<5	<5	<5	Comp	EPA 625
☐	Di-n-butyl phthalate	84742	<5	<5	<5	<5	Comp	EPA 625
☐	Di-n-octyl phthalate	117840	<5	<5	<5	<5	Comp	EPA 625
☐	Dibenzo(a,h) anthracene	53703	<5	<5	<5	<5	Comp	EPA 625
☐	1,2-dichlorobenzene	95501	<5	<5	<5	<5	Comp	EPA 625
☐	1,3-dichlorobenzene	541731	<5	<5	<5	<5	Comp	EPA 625
☐	1,4-dichlorobenzene	106467	<5	<5	<5	<5	Comp	EPA 625
☐	3,3-dichlorobenzidine	91941	<20	<20	<20	<20	Comp	EPA 625
☐	Diethyl Phthalate	84662	<5	<5	<5	<5	Comp	EPA 625
☐	Dimethyl Phthalate	131113	<5	<5	<5	<5	Comp	EPA 625
☐	2,4-dinitrotoluene	121142	<5	<5	<5	<5	Comp	EPA 625
☐	2,6-dinitrotoluene	606201	<5	<5	<5	<5	Comp	EPA 625
☐	1,2-diphenylhydrazine	122667	<5	<5	<5	<5	Comp	EPA 625
☐	Fluoranthene	206440	<5	<5	<5	<5	Comp	EPA 625
☐	Fluorene	86737	<5	<5	<5	<5	Comp	EPA 625
☐	Hexachlorobenzene	118741	<5	<5	<5	<5	Comp	EPA 625
☐	Hexachlorobutadiene	87683	<5	<5	<5	<5	Comp	EPA 625
☐	Hexachlorocyclopentadiene	77474	<5	<5	<5	<5	Comp	EPA 625
☐	Hexachloroethane	67721	<5	<5	<5	<5	Comp	EPA 625
☐	Indeno(1,2,3-cd) pyrene	193395	<5	<5	<5	<5	Comp	EPA 625
☐	Isophorone	78591	<5	<5	<5	<5	Comp	EPA 625
☐	Naphthalene	91203	<5	<5	<5	<5	Comp	EPA 625
☐	Nitrobenzene	98953	<5	<5	<5	<5	Comp	EPA 625
☐	N-nitrosodi-n-propylamine	None	<5	<5	<5	<5	Comp	EPA 625
☐	N-nitrosodimethylamine	62759	<5	<5	<5	<5	Comp	EPA 625
☐	N-nitrosodiphenylamine	86306	<5	<5	<5	<5	Comp	EPA 625
☐	Phenanthrene	85018	<5	<5	<5	<5	Comp	EPA 625
☐	Pyrene	129000	<5	<5	<5	<5	Comp	EPA 625
☐	1,2,4-trichlorobenzene	120821	<5	<5	<5	<5	Comp	EPA 625

C. Collection System Information

FACILITY NAME Cadillac WWTP	NPDES PERMIT NUMBER MI0020257
--------------------------------	----------------------------------

Complete this item if there are outfalls at the treatment facility or along the combined sewer collection system from which discharges of untreated or partially-treated wastewater occur. Update the information in items A, B, and C below if it has changed since the last Application submission.

- A. Estimate the percentage of the collection system that is combined: 0 %
- B. System Map. Provide a map that shows all Combined Sewer Outfall discharge points.
- C. System Diagram. Provide a diagram, in the above map or on a separate drawing, of the combined sewer collection system. Include the locations of major trunk line sewers, both combined and separate sanitary; the locations of points where separate sanitary sewers feed into the combined sewer system; the locations of in-line and off-line storage structures; locations of flow regulating devices, and the locations of pump stations.

Identify the outfall(s) from your current permit by number (e.g., 001, 002). Provide the current status of the outfall (i.e. active, inactive, bulk headed, removed) and indicate if continued authorization is required. Attach specific outfall location and discharge information for all outfalls not previously identified.

EQP 4659-C (Rev. 3/2011)

Michigan Department of Environmental Quality – Water Resources Division
WASTEWATER DISCHARGE PERMIT APPLICATION
SECTION IV – Storm Water

PLEASE TYPE OR PRINT

FACILITY NAME
Cadillac WWTP

NPDES PERMIT NUMBER
MI0020257

1. STORM WATER DISCHARGES

Facilities that discharge storm water must provide the following information. (Note: The following discharges are also covered by a storm water authorization, provided they are addressed in the facility's Storm Water Pollution Prevention Plan [SWPPP]): discharges from fire hydrant flushing; potable water sources, including water line flushing; fire system test water; irrigation drainage; lawn watering; routine building wash down which does not use detergents or other compounds; pavement wash waters where spills or leaks of toxic or hazardous materials have not occurred (unless all spilled material has been removed) and where detergents are not used; air conditioning condensate; springs; uncontaminated ground water; and foundation or footing drains where flows are not contaminated with process materials such as solvents. **Unless otherwise specified, answer the following questions.**

A. Is the storm water from this facility discharged to the waters of the state either directly or through another conveyance? Note: If storm water is discharged to a municipal wastewater treatment system or a privately-owned activated sludge treatment system, check the "No" box.

- ☒ Yes.
☐ No.

B. Is the facility identified in this Application primarily engaged in an "industrial activity" as defined in 40 CFR 122.26(b)(14)?

- ☒ Yes.
☐ No.

C. Are there any industrial activities or materials exposed to storm water at this facility? Storm water discharge requirements may be excluded from an NPDES Permit when there are no industrial activities or materials exposed to storm water. To qualify, the applicant shall certify that the facility has met all the eligibility requirements to claim a condition of "no exposure." These requirements are found in the No Exposure Certification Form in the Appendix. This form is also available on the DEQ's Internet Page. To access the form, go to <http://www.michigan.gov/deq>. In the left column, click on WATER, click on Surface Water, click on Storm Water, in the middle column click on Industrial Program, then click on No Exposure Certification.

- ☐ Yes.
☒ No. Complete the No Exposure Certification Form in the Appendix and submit it with this Application.

D. Does this facility have a current and up-to-date SWPPP?

- ☒ Yes.
☐ No. **Note: The applicant must complete this program element to receive storm water discharge authorization.**

E. Has the facility implemented the nonstructural controls described in the SWPPP?

- ☒ Yes.
☐ No. **Note: The applicant must complete this program element to receive storm water discharge authorization.**

F. Have all the structural controls described in the SWPPP been constructed and put into operation?

- ☐ Yes.
☐ No. **Note: The applicant must complete this program element to receive storm water discharge authorization.**

G. Does this facility have a certified industrial storm water operator who supervises the facility's storm water treatment and control measures included in the SWPPP?

- ☒ Yes. Jeffrey M. Dietlin 02413
Storm Water Operator Name Certification Number
☐ No. **Note: The applicant must complete this program element to receive storm water discharge authorization.**

H. Is any of the storm water discharged from (check all that apply):

- ☐ Secondary containment structures that are required by state or federal law. On a separate page, provide a list of the materials that are stored in this area.
☐ Areas identified on Michigan's list of Sites of Environmental Contamination, pursuant to the Natural Resources and Environmental Protection Act, 1994 PA 451, as amended, Part 201 (formerly 307).

I. The storm water from this facility discharges to the following receiving water(s): Clam River

Applicants should provide any sample data taken of the storm water discharge as an attachment. To submit additional information, see Page ii, Item 3.